



SYMBIOSIS
SCHOOL OF ECONOMICS

**Time Series Forecasting of Stock Prices: Using
ARMA, ARCH, and GARCH Models for
Adani Green Energy**

B.Sc. (Economics) Honours (2022-25)

Intermediary Econometrics

Group: 6

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1. Abstract

This paper carries out time series analysis of the stock prices of Adani Green Energy Ltd. This analysis employed the ARMA, ARCH, and GARCH models in forecasting the price volatility dated June 2018 to October 2024. AGEL is one of the major renewable energy companies in India whose stock prices have undergone tremendous volatility due to their dynamics and policy changes in the market. The ARMA model picks up the short-run fluctuations in the stock return, while the ARCH and GARCH pick up volatility patterns of clustering, which have periods of high volatility followed by similar periods. The results of the study imply persistence but mean reversion in the volatility of stock prices, which eventually stabilize.

Keywords: ARMA, ARCH, GARCH, Forecasting, Volatility, Stock Return

2. Introduction

Over the past few years, there has been a significant increase in the demand for renewable energy probably because of the increasing effects of climate change. India being one of the biggest consumers of energy in the world has significantly improved on the use of renewable energy. (Singh, n.d.) The main driver of this change is Adani Green Energy Ltd. (AGEL), the

renewable arm of the Indian multinational conglomerate Adani Group (CNBC, 2024). Being a solar and wind energy project developer, AGEL occupies a strategic position in helping the country realize its enhanced green energy goals.

Since AGEL's listing on the stock exchange in June 2018, the entity has gained much attention from investors as well as policymakers indicating its strategic importance to the overall Energy profile of India. As it is seen for the investors, it is necessary to clarify the behaviour of stock prices especially in the high-growth industries such as renewable energy. It is used to forecast more stock prices movements when specific historical patterns of time and volatility have been established.

Analysing the data obtained since the stock listing of AGEL in June 2018, this report employs ARMA (Autoregressive Moving Average), ARCH (Autoregressive Conditional Heteroskedasticity), and GARCH (Generalized ARCH) models to predict volatility in the firm's stock prices (Harvey & Chakravarty, 2008). The following models have been used in the report to provide an understanding of how AGEL stock price has changed over time, especially in very relevant market and policy conditions.

Fluctuations in stock price are not mere points on monetary fluctuations, but rather an indication of how investors feel about the potential risks ...and rewards of a company. Issues such as this are vital in industries that experience volatilities in terms of technology, regulations, and policies such as renewable energy. Of all the models whose descriptive statistics and yield are examined in this paper, the ARMA, ARCH, and GARCH models are particularly suited to capture the time-dependency nature of the stock prices, in the sense of isolating out periods of relative stability and volatile movements that may well be related to important events in the financial market or policy shocks (Alm et al., 2013).

Adopting the key features of Adani Green Energy allows for adopting a view of the main objectives of Indian renewable energy legislation in the financial context (Gupta & Jindal, 2024). The sector has gained momentum since 2019 due to enhancement policies like the National Solar Mission and government goals toward reaching 450 GW of RE capacity in 2030. The continued development in the number of projects being undertaken by AGEL makes it an ideal test subject concerning the fluctuations of its stock price and its volatility in the renewable energy market.

The report begins by outlining the background to the Indian renewables market and AGEL's contribution to supporting that market's development. This research contributes in the literature which emphasizes that investors and policymakers can assess financial market dynamics during India's rapid renewable energy growth by utilizing models like ARMA, ARCH, and GARCH to understand stock price volatility and the impact of policy shifts on companies such as Adani Green Energy.

3. Research Objectives

- To analyse the volatility of Adani Green Energy Ltd. in terms of their stock prices using time series models.
- Use of ARMA, ARCH, and GARCH models to forecast possible future movements of the stock price.

4. Literature Review

In the past few years, renewable power generation has developed notably, as well as experiments with climate change and the search for innovative solutions. India, being one of the largest energy consumers, has welcomed this shift, setting renewable energy as part of its national interest. Adani Green Energy (AGEL) is one of the

prominent entities that have speedily transformed into one of India's largest renewable energy firms. AGEL mainly engages in the large-scale solar and wind power sector, which has enormously contributed to India's renewable energy target. AGEL went live on NASDAQ in June 2018 and quickly became an investor and policymaker's preferred because of its central position in India's renewable energy sector (Ekong & Onye, 2017). Specifically, it is necessary to reveal the trends, fluctuations in the position of the companies' stock, and oscillations of Adani Green Energy, which contain its strengths and weaknesses to assess the sustainability of renewable energy and potential risks for investors. In this case, time series analysis is one of the best stock price and market volatility forecasting methods (Kirchgassner et al., 2012). This study uses three models, ARMA (Auto Regressive Moving Average), ARCH (Autoregressive Conditional Heteroskedasticity), and GARCH (Generalized ARCH), as models to predict the volatility surrounding AGEL's stock prices in its operating period since its listing in June 2018 (Tripathy & Rehman, 2013).

The ARMA model is a powerful tool in time series forecasting that combines two components: auto regressive (AR) and moving average (MA). The AR part of the

model analyses the effects of a time series' previous values (lags), while the MA part analyses the past errors (residuals) or shocks. Combined, they assist in identifying future values of a particular stock under evaluation using its records. The reason why ARMA is useful in case of Adani Green Energy is that helps in finding out the correlations and thus predict the short-term movements of the prices of the stocks. ARMA is especially useful if there is Autoregressive Moving Average in the time series data and such data patterns are common in stock prices.

In finance, volatility is the measure of the relative amount of change in the stock price during a specified period, whereas variance is stochastic measure of this volatility. The consistently prescribes that high volatility events are followed by more volatility events in stock markets leading to volatility clusters. This is where models known as ARCH (Autoregressive Conditional Heteroskedasticity) and GARCH (Generalized ARCH) enter the picture. The ARCH model presented by (Engle, 1982) is in fact developed to capture the evolving variance. It presumes that frequent fluctuations present in today's financial market have resulted from some other periods of volatility (Bollerslev, 1986). In layman language, if the stock has had a big movement in terms of price (either up or

down) in the past, ARCH model aids to predict that more than usual volatility is still in store for the stock. This clustering of volatility can be useful where the operating environment can dramatically affect its business performance and stock price movements; some examples being government policies, global energy prices, and regional sentiments etc.

GARCH model goes a step further than ARCH by considering not only past shocks but the past variance as well. Most of all, it is serving long-term forecasting needs well because it captures roll force volatility effectively (Gabriel, 2012). In the case of Adani Green Energy, in the industry where policy and/or environmental shifts can exert permanent effects on the shares' production, GARCH gives better predictive capability of the shares' future volatility. It helps investors know not just the market movement in the wake of market information but also possibly continuing market fluctuations. Adani Green Energy is highly relevant for study because its developments offer interesting insights regarding the transition to low-carbon economies as well as the role of renewable energy in India's electricity supply. Continuing the same note, Adani Green Energy is much more than just another renewable energy company; it defines India's green energy playbook (Garg,

2012). Subsequently, since its inception in 2015, the company has grown quickly due to an increasing number of major solar and wind projects. AGEL has committed to its strategic vision for 2030, which is to add 45 GW of renewable energy capacity, which fits with the overall objective of achieving 450 GW of renewable energy capacity in India by the same year. This makes AGEL one of the most important companies involved in the progress of renewable energy resources in the country.

AGEL achieved huge attention after becoming a listed company on stock exchange owing to its big-ticket projects and the strategic sponsorship of the Adani Group, one of the largest business conglomerates in India. Analysing the fluctuation in the company stocks, it can extrapolate that the performance of the company can be volatile due to factors which include growth and expansion policies done by the government, general outlook on energy in the global market, technological developments a ready in renewable energy (Tutak et al., 2021). Due to this volatility, Adani Green Energy provides a perfect example as to how financial markets respond to news in the renewable energy sector. A critical concept to learn in volatility analysis is the importance of forecasting. volatility represents a critical concept to investors

since it helps them to address risk. Volatility is important for dealing with investment plans, portfolio, and managing risks. Using the ARMA, ARCH, and GARCH models, we can make AGEL's actual stock price volatility that will present clearer picture on how the stock respond to both the market and the company's activeness. Hence, this goes a long way in showing that modelling of volatility transcends the mere explanation of fluctuation of stock price. It provides additional understanding of risk and uncertainty based on perceptions observed on the market. For instance, variations in political science in relation to the renewable energy sector ranging from subsidies to regulatory measures will trigger volatility (Bhowmik et al., 2016). Also, the international changes in price levels for energy and environmental regulatory measures, as well as the technological advancements affecting the renewable energy segment, add to fluctuations in investors' appetite and stock performance.

In 2019, India launched several policies to develop renewable energy, such as the National Solar Mission and the renewed goal to reach 450GW in Renewable Energy by 2030 (Alvarez et al., 2014). These policies have catalysed more investments in renewable energy systems thereby having influences companies like Adani Green

Energy. On the same note, through an analysis of fluctuations in the AGEL stock prices, this paper will explain how such policies affect financial markets. This report not only aims to determine AGEL's forecasted stock price volatility, but also analyses the overall impact of India's renewable energy legislation on the market. By applying the ARMA, ARCH, and GARCH models to stock price data from June 2018 onwards, we gain valuable insights into volatility patterns, offering useful information for investors and policymakers who are shaping the future of renewable energy in India. Therefore, micro stage analysis of Adani Green Energy's stock price and volatility is relevant in the analysis of this company since it is the forefront of the renewable energy production in India out of all the companies included in this research study. For this reason, the application of ARMA, ARCH, and GARCH models offers a comprehensive setting in forecasting stock price movements in a volatile industry. Due to this we use the monthly stock price data of AGEL to understand the volatility and prospects of RE sector and guide other investors willing to invest in the renewable energy sector, while also assisting policymakers in implementing green energy plans for India.

5. Research Methodology

The paper aims to utilize the time series models such as ARMA, ARCH and GARCH for the volatility forecasting of the Adani Green Energy stock price. The method utilizes secondary data and

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t$$

employs techniques of econometric analysis on the stock's return volatility over a period. All analyses were conducted using EViews, statistical software for time series analysis, especially for the ARMA, ARCH, and GARCH models.

5.1 Data Collection

The data used for the analysis are historical stock prices of Adani Green Energy, sourced on daily basis from Investing.com. An analysis period from June 2018 to October 2024 was chosen to forecast the volatility phenomenon. The data was then transformed from daily stock prices into logarithmic returns so that the stability of the data is ensured and potential heteroscedasticity and non-stationarity in time series are reduced.

5.2 Model Selection and Specification

5.2.1 ARMA (Autoregressive Moving Average) Model:

The ARMA model integrated both the autoregressive and moving average components. It accounted for the current value of a time series using both its past values and past error terms. The general form of the ARMA(p,q) model is:

We estimated three equations:

- AR(1) MA(1)
- AR(1) MA(0)
- AR(0) MA(1)

The ARMA (1,0) equation should the most significant results.

5.2.2 Autoregressive Conditional Heteroskedasticity (ARCH) Model:

In the case of time series like stock prices, volatility frequently turns out to change overtime. To consider this, we employed the ARCH (Autoregressive Conditional Heteroskedasticity) model. In an ARCH model, the current period's volatility depends on past squared errors, which represent past periods of high or low volatility. The generalized form of an ARCH model is:

- Mean Equation:

$$Y_t = c + \phi_1 Y_{t-1} + \epsilon_t$$

- Variance Equation:

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2$$

5.3.3 Generalized Autoregressive Conditional Heteroskedasticity (GARCH) Model:

The GARCH model extends the ARCH model. It allows for the fact that volatility itself is a time series process, which is good for modelling long-term volatility. The general form of an GARCH model is:

$$\epsilon_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

6. Results and Discussion

6.1 Volatility Clustering

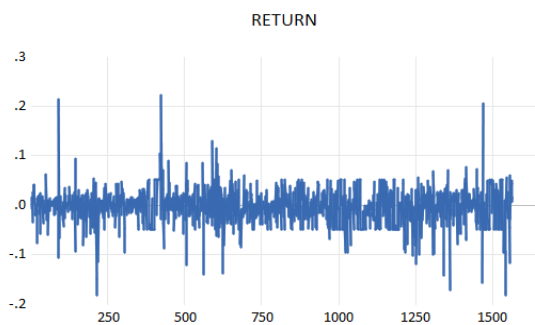


Figure. 1 Volatility clustering of daily returns of Adani green energy

The time series plot of the stock returns of Adani Green Energy has extreme volatility clustering as shown in Figure 1. Positive and negative spikes are observed, showing sharp price movements at certain periods of times, which can be assessed because of market events.

6.2 Augmented Dickey Fuller Test

The ADF Test is used to check the stationarity of the dataset. We create:

H_0 : There is Unit Root.

H_1 : There is no Unit Root.

Null Hypothesis: RETURN has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=23)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-31.39286	0.0000
Test critical values: 1% level	-3.434328	
5% level	-2.863184	
10% level	-2.567693	

Figure 2. ADF Test for Stationarity

Since the p-value is less than 0.05, the time series of returns for Adani Green Energy has a unit root, meaning the series is non-stationary.

6.3 ARMA Model

This output presents the estimation of an ARMA(1,0) model for Adani Green Energy's stock returns. Specifically, it uses one autoregressive term AR(1) and no moving average terms (MA).

Dependent Variable: RETURN
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 10/12/24 Time: 16:56
Sample: 2 1565
Included observations: 1564
Convergence achieved after 25 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.002553	0.001144	-2.231577	0.0258
AR(1)	0.225136	0.013534	16.63528	0.0000
SIGMASQ	0.001185	2.11E-05	56.18160	0.0000
R-squared	0.050678	Mean dependent var		-0.002562
Adjusted R-squared	0.049461	S.D. dependent var		0.035347
S.E. of regression	0.034462	Akaike info criterion		-3.895998
Sum squared resid	1.853837	Schwarz criterion		-3.885726
Log likelihood	3049.670	Hannan-Quinn criter.		-3.892179
F-statistic	41.66549	Durbin-Watson stat		2.008421
Prob(F-statistic)	0.000000			
Inverted AR Roots	.23			

Figure 3. ARMA Model

From the results, the estimated equation is:
 $\text{Return}_t = -0.002553 + 0.225136 * \text{Return}_{t-1} + \epsilon_t$

The constant is -0.002553, with a p-value of 0.0258 which is less than 0.05.

This means that, on average, the stock's return tends to have a very small drift which is negative over time.

The AR(1) coefficient is 0.225136, with a strong t-statistic and a p-value of 0.0000, that is highly significant. This tells us that approximately 22.5% of yesterday's return determines today's return. If the stock went up yesterday, then around 22.5% of that movement spills over into today's return.

The value of adjusted R-squared at 4.94% shows relationship in the stock return. Though it captures some relation between past and current returns. This simply implies that the model will account for only a small part of the return variation while most of the stock movement remains due to events outside the model such as market events and volatility clustering.

6.3.1 Diagnostic for residuals

After the estimation step of ARMA Model, we check whether the residual series is at stationary or not.

Null Hypothesis: RESID02 has a unit root Exogenous: Constant Lag Length: 0 (Automatic - based on SIC, maxlag=23)		
	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-39.67669	0.0000
Test critical values: 1% level	-3.434328	
5% level	-2.863184	
10% level	-2.567693	

Figure 4. Stationarity for residuals

Residuals were found stationary at level since p is less than 0.05, rejecting the null hypothesis.

6.4 ARCH-GARCH Model

We test for heteroskedasticity, to verify that the variance of the residuals is changing. In our model, heteroskedasticity holds, this justifies the use of ARCH and GARCH model which are designed to capture volatility.

Dependent Variable: RETURN
Method: MLARCH - Normal distribution (BFGS / Marquardt steps)
Date: 10/12/24 Time: 16:57
Sample (adjusted): 3 1565
Included observations: 1563 after adjustments
Convergence achieved after 26 iterations
Coefficient covariance computed using outer product of gradients
Presample variance: backcast (parameter = 0.7)
GARCH = C(3) + C(4)*RESID(-1)^2 + C(5)*GARCH(-1)

Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	-0.001972	0.000857	-2.301981	0.0213
AR(1)	0.174710	0.027787	6.287528	0.0000

Variance Equation

C	0.000307	2.27E-05	13.49789	0.0000
RESID(-1)^2	0.354703	0.031576	11.23328	0.0000
GARCH(-1)	0.413982	0.031699	13.05957	0.0000

R-squared	0.047941	Mean dependent var	-0.002559
Adjusted R-squared	0.047331	S.D. dependent var	0.035358
S.E. of regression	0.034511	Akaike info criterion	-4.068012
Sum squared resid	1.859165	Schwarz criterion	-4.050883
Log likelihood	3184.151	Hannan-Quinn criter.	-4.061644
Durbin-Watson stat	1.904068		

Inverted AR Roots .17

Figure 5. ARCH-GARCH Model

Estimated Mean Equation (ARCH)

$$\text{Return}_t = -0.001972 + 0.174710 * \text{Return}_{t-1} + \epsilon_t$$

The outputs of ARCH model on stock return shows that the constant is -0.001972 with a p-value of 0.0213 which indicates that constant is significant at 5% level. The AR(1) coefficient is 0.174710 with a p-value of 0.0000. It indicates that the return series is autocorrelated, meaning that the

current return is positively influenced by the return from the previous period.

Estimated Variance Equation (GARCH)

$$\sigma_t^2 = 0.000307 + 0.354703 * \epsilon_{t-1}^2 + 0.413982 * \sigma_{t-1}^2$$

The variance equation shows the volatility in the model. The constant is 0.000307 with a p-value of 0.0000 which indicates long term variance. ARCH coefficient is 0.354703 with a p-value of 0.0000 represents the ARCH effect, where the current variance is a function of the square of the previous period's residual. Thus, a large value of this coefficient points towards a strong, immediate impact of past volatility. The GARCH coefficient is 0.413982 with a p-value of 0.0000, signifies that the time series has volatility clustering, that is, periods of high volatility are followed more often by periods of high volatility.

The sum of ARCH (0.354703) and GARCH (0.413982) coefficients is 0.768685, which is less than 1. This signifies that volatility can increase for a period due to market uncertainty and it will eventually return back to normal levels.

6.5 Forecasting

We begin with an initial sample of 1,565 observations of stock return that captures the daily movement of the Adani Green Energy stock for a decent amount of time.

To further extend the analysis and in turn make a prediction about its future performance, we increase the sample size from 1,565 observations to 1,645 observations by adding another 80 data points based on the historical data points. This model based on past values is used in the forecast. It predicts future returns by the trends in the data presented.

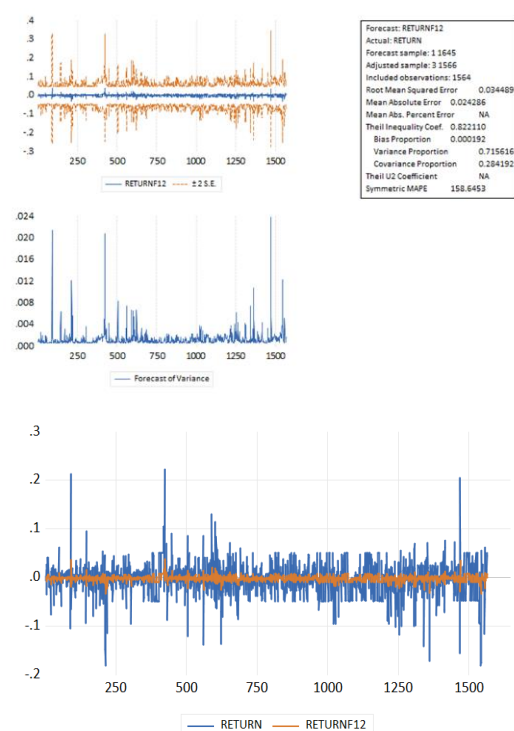


Figure. 6 Forecasting

The top graph indicates the actual returns in blue, which represent the range within which we will expect most of the forecasted values most of the time. Most of the time, the actual values lie within those bands, which would mean that the model makes good predictions under normal market conditions. However, there are several

spikes where the actual returns cross these bands, thereby implying that the model cannot reflect extreme movements or sudden market shifts. Such outliers might have resulted from potentially unexpected events such as news or policy changes and hence may indicate that although the model works for daily fluctuations, it is not so effective when there are periods of high volatility.

The bottom graph illustrates the forecasted variance (volatility) which means that it is the measure of how much the returns are expected to vary from their average during certain periods of time. The graph shows that at some points in time, the variance explodes. This is expressed as episodes in time, meaning episodes of increased market activity or instability. These spikes present a higher risk at given times, perhaps due to factors such as shifts in market sentiment or changes in economic conditions. This pattern reveals that the model is doing its best to capture history but fails to forecast rapid changes in the series.

7. Conclusion

The paper analyses the stock price of AGEL so estimates the future volatility using time series models, specifically the ARMA, ARCH and GARCH. Volatility clustering is

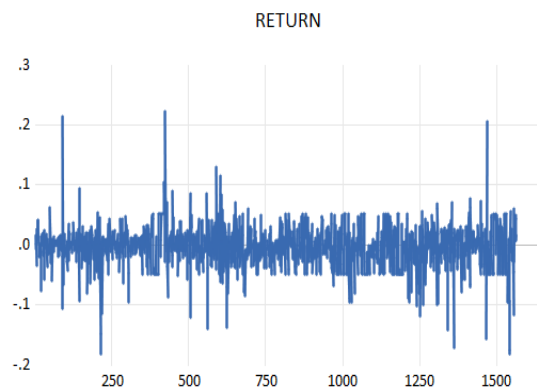
identified as prevalent in volatility patterns, with high volatility being preceded by similar periods, and low volatility, the same way around. Using ARMA modelling, we noticed that around 22.5% of returns from the previous period influences current returns, meaning at least some predictability over short-term horizons. ARCH and GARCH models prove that volatility is persistent yet returns to normal over time. The sum of the coefficients of ARCH and GARCH is less than 1, thus explaining that stock prices may remain unstable over a long period but return to a normal level over time. Moreover, policy related changes in India, especially on renewable energy initiatives, equally have a significant impact on the movements of stocks. The paper brings risks regarding the stock of AGEL into perspective through predictions of volatility and can help investors make decisions based on that. The policymakers will be well-informed to predict the market outcome of a change in regulation. The models were effective in capturing the general trends as well as the inherent volatility of the AGEL stock but not necessarily extreme market shocks; while models may require further refinement when dealing with abrupt shifts in markets.

8. References

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9. Appendices

A. Volatility Clustering for Stock Return



B. Augmented Dickey Fuller Test

Null Hypothesis: RETURN has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=23)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-31.39286	0.0000
Test critical values:		
1% level	-3.434328	
5% level	-2.863184	
10% level	-2.567693	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation

Dependent Variable: D(RETURN)

Method: Least Squares

Date: 10/12/24 Time: 16:51

Sample (adjusted): 3 1565

Included observations: 1563 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
RETURN(-1)	-0.774722	0.024678	-31.39286	0.0000
C	-0.001975	0.000874	-2.259237	0.0240
R-squared	0.387005	Mean dependent var	3.64E-05	
Adjusted R-squared	0.386612	S.D. dependent var	0.044001	
S.E. of regression	0.034461	Akaike info criterion	-3.896677	
Sum squared resid	1.853822	Schwarz criterion	-3.889825	
Log likelihood	3047.253	Hannan-Quinn criter.	-3.894130	
F-statistic	985.5118	Durbin-Watson stat	2.008679	
Prob(F-statistic)	0.000000			

C. Correlogram

Date: 10/12/24 Time: 16:52

Sample (adjusted): 2 1565

Included observations: 1564 after adjustments

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.225	0.225	79.298	0.000
		2	0.071	0.021	87.157	0.000
		3	0.079	0.062	97.013	0.000
		4	0.072	0.042	105.11	0.000
		5	0.077	0.050	114.43	0.000
		6	0.070	0.038	122.18	0.000
		7	0.044	0.012	125.27	0.000
		8	0.057	0.035	130.34	0.000
		9	0.011	-0.022	130.52	0.000
		10	0.056	0.048	135.47	0.000
		11	0.055	0.023	140.23	0.000
		12	0.044	0.019	143.32	0.000
		13	0.042	0.018	146.15	0.000
		14	-0.018	-0.047	146.69	0.000
		15	0.003	0.004	146.70	0.000
		16	-0.006	-0.022	146.77	0.000
		17	-0.045	-0.048	149.97	0.000
		18	-0.032	-0.022	151.63	0.000
		19	-0.025	-0.013	152.58	0.000
		20	0.021	0.037	153.30	0.000
		21	-0.031	-0.041	154.83	0.000
		22	-0.019	0.004	155.43	0.000
		23	-0.040	-0.038	158.02	0.000
		24	0.008	0.035	158.12	0.000
		25	-0.016	-0.016	158.51	0.000
		26	-0.042	-0.031	161.32	0.000
		27	-0.024	0.004	162.27	0.000
		28	-0.030	-0.019	163.71	0.000
		29	-0.024	0.004	164.65	0.000
		30	-0.049	-0.041	168.46	0.000
		31	-0.003	0.028	168.48	0.000
		32	-0.011	-0.010	168.68	0.000
		33	-0.017	-0.001	169.12	0.000
		34	-0.034	-0.020	170.96	0.000
		35	-0.009	0.004	171.08	0.000
		36	-0.024	-0.013	171.97	0.000

D. ARMA Model

Dependent Variable: RETURN

Method: ARMA Maximum Likelihood (OPG - BHHH)

Date: 10/12/24 Time: 16:56

Sample: 2 1565

Included observations: 1564

Convergence achieved after 25 iterations

Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.002553	0.001144	-2.231577	0.0258
AR(1)	0.225136	0.013534	16.63528	0.0000
SIGMASQ	0.001185	2.11E-05	56.18160	0.0000
R-squared	0.050678	Mean dependent var	-0.002562	
Adjusted R-squared	0.049461	S.D. dependent var	0.035347	
S.E. of regression	0.034462	Akaike info criterion	-3.895998	
Sum squared resid	1.853837	Schwarz criterion	-3.885726	
Log likelihood	3049.670	Hannan-Quinn criter.	-3.892179	
F-statistic	41.66549	Durbin-Watson stat	2.008421	
Prob(F-statistic)	0.000000			
Inverted AR Roots	.23			

E. Residual Series

Null Hypothesis: RESID02 has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=23)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-39.67669	0.0000
Test critical values: 1% level	-3.434328	
5% level	-2.863184	
10% level	-2.567693	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(RESID02)
Method: Least Squares
Date: 10/12/24 Time: 16:56
Sample (adjusted): 3 1565
Included observations: 1563 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
RESID02(-1)	-1.004701	0.025322	-39.67669	0.0000
C	3.01E-06	0.000872	0.003452	0.9972
R-squared	0.502111	Mean dependent var	2.95E-05	
Adjusted R-squared	0.501792	S.D. dependent var	0.048823	
S.E. of regression	0.034461	Akaike info criterion	-3.896699	
Sum squared resid	1.853781	Schwarz criterion	-3.889848	
Log likelihood	3047.270	Hannan-Quinn criter.	-3.894152	
F-statistic	1574.239	Durbin-Watson stat	1.998924	
Prob(F-statistic)	0.000000			

F. Heteroskedasticity

Heteroskedasticity Test: ARCH

F-statistic	185.0797	Prob. F(1,1561)	0.0000
Obs*R-squared	165.6738	Prob. Chi-Square(1)	0.0000

Test Equation:
Dependent Variable: RESID*2
Method: Least Squares
Date: 10/12/24 Time: 16:57
Sample (adjusted): 3 1565
Included observations: 1563 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000800	8.69E-05	9.214547	0.0000
RESID*2(-1)	0.325563	0.023931	13.60440	0.0000
R-squared	0.105997	Mean dependent var	0.001186	
Adjusted R-squared	0.105425	S.D. dependent var	0.003431	
S.E. of regression	0.003245	Akaike info criterion	-8.621825	
Sum squared resid	0.016442	Schwarz criterion	-8.614973	
Log likelihood	6739.956	Hannan-Quinn criter.	-8.619278	
F-statistic	185.0797	Durbin-Watson stat	1.951798	
Prob(F-statistic)	0.000000			

G. ARCH & GARCH Model

Dependent Variable: RETURN
Method: ML ARCH - Normal distribution (BFGS / Marquardt steps)
Date: 10/12/24 Time: 16:57
Sample (adjusted): 3 1565
Included observations: 1563 after adjustments
Convergence achieved after 26 iterations
Coefficient covariance computed using outer product of gradients
Presample variance: backcast (parameter = 0.7)
GARCH = C(3) + C(4)*RESID(-1)*2 + C(5)*GARCH(-1)

Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	-0.001972	0.000857	-2.301981	0.0213
AR(1)	0.174710	0.027787	6.287528	0.0000

Variance Equation				
C	0.000307	2.27E-05	13.49789	0.0000
RESID(-1)*2	0.354703	0.031576	11.23328	0.0000
GARCH(-1)	0.413982	0.031699	13.05957	0.0000

R-squared	0.047941	Mean dependent var	-0.002559
Adjusted R-squared	0.047331	S.D. dependent var	0.035358
S.E. of regression	0.034511	Akaike info criterion	-4.068012
Sum squared resid	1.859165	Schwarz criterion	-4.050883
Log likelihood	3184.151	Hannan-Quinn criter.	-4.061644
Durbin-Watson stat	1.904068		

Inverted AR Roots .17